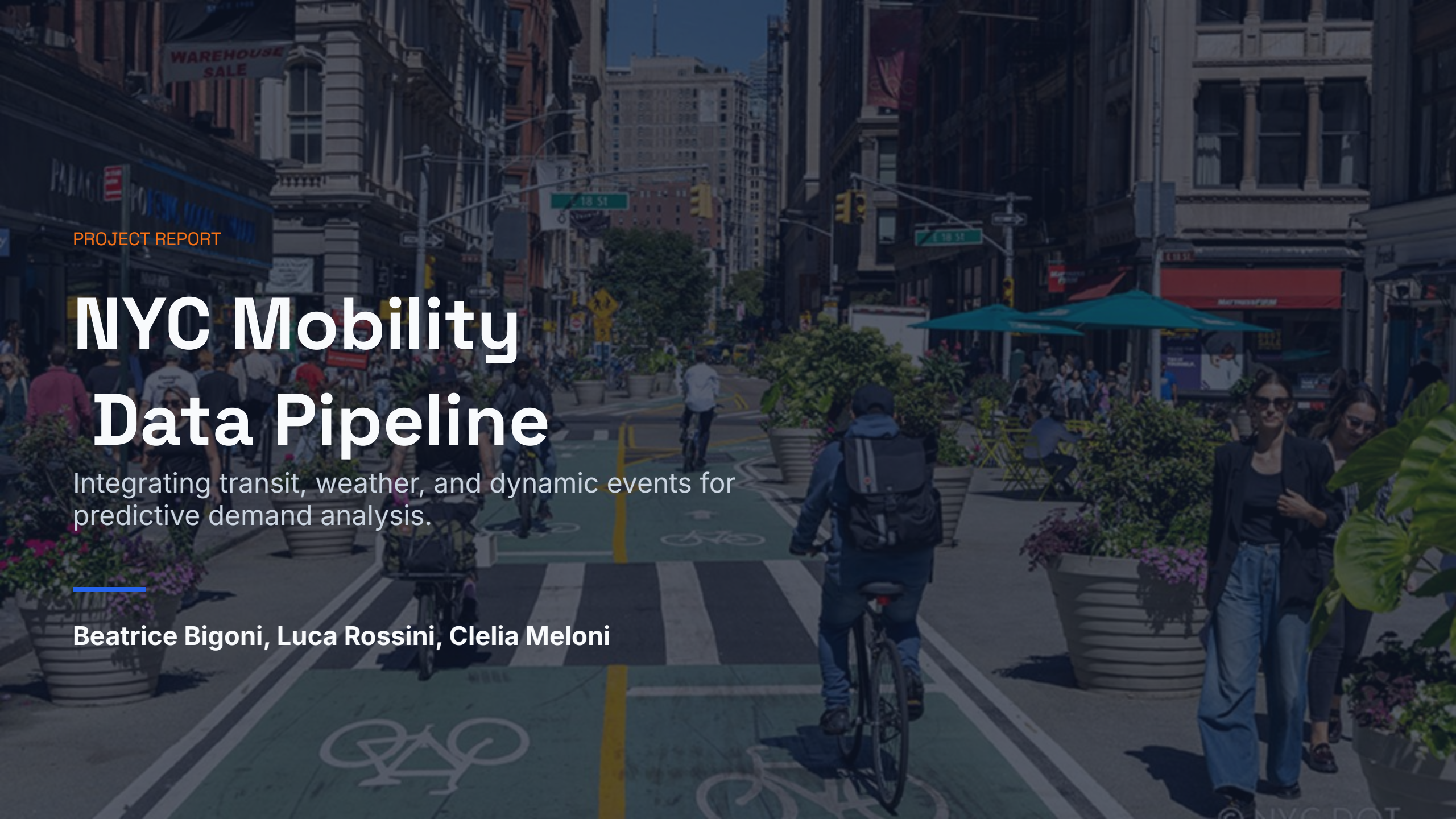


PROJECT REPORT

NYC Mobility Data Pipeline

Integrating transit, weather, and dynamic events for predictive demand analysis.

Beatrice Bigoni, Luca Rossini, Clelia Meloni

1. INTRODUCTION

Urban Mobility Dynamics

Complex Logistics

Citi Bike integration within urban connective tissue.

Dynamic Variables

Adverse weather acts as a usage disincentive.

Major Events

Madison Square Garden activities stimulate demand.



2. OBJECTIVES

Research Questions

Primary Focus

Meteorological Influence

Measuring precipitation and temperature impact on hourly trip volumes.

Secondary Focus

Sporting Events

Quantifying demand alterations during Knicks home games against baseline.

3.0 & 4.1 CITIBIKE INTEGRATION

Transactional Log Processing

Batch Processing

Iterative CSV file ingestion.

Memory Optimization

Retained only temporal and user attributes.

Physiological Filter

Trips restricted to 1–180 minute durations.

3.1 & 4.2 WEATHER INTEGRATION

Open-Meteo REST API

Persistent Caching
Local SQLite storage prevents redundant network calls.

Retry Logic
Automatic 5-attempt retry with exponential backoff.

Timezone Alignment
UTC timestamps converted to NY naive datetime format.

3.2 & 4.3 EVENT INTEGRATION

Knicks Dynamic Web Scraping

Selenium Automation

Bypassed static limitations to handle cookie banners and asynchronous HTML tables.

Engineered strict "Home vs Away" logic to isolate Madison Square Garden spatial impact.



5.1 & 5.2 DATA INTEGRATION

Granularity Standardization



Hourly Buckets

Uniform timeline aggregation across all datasets.





Primary Key

Deterministic YYYYMMDDHH identifier generation.

5.3 INTEGRATION

Investigation of Correspondences

100%

Weather Overlap

Zero missing temporal gaps.

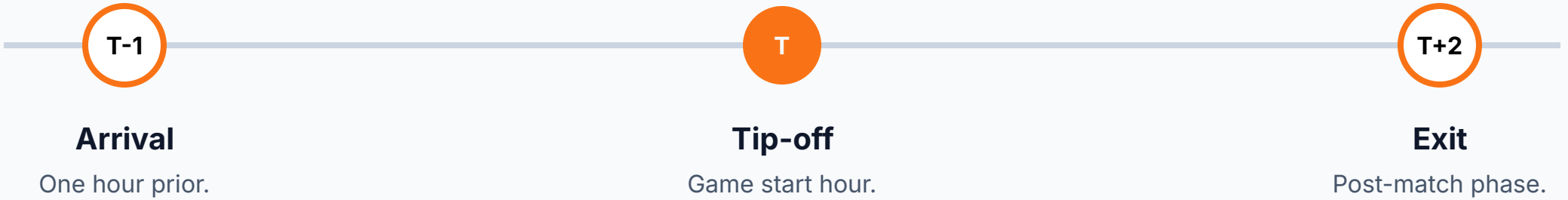
Valid

Event Alignment

All games mapped securely.

5.4 CONSOLIDATION

Temporal Window Expansion



5.5 TARGET VARIABLE CONSTRUCTION

Demand Classification

Class 0

Low Demand

Below 25th percentile.

Class 1

Medium Demand

Interquartile range.

Class 2

High Demand

Above 75th percentile.

6. DATA QUALITY

Continuous Quality Assessment

Fixing at Source

Extended API acquisition window to resolve UTC blind spots.

Syntactic Accuracy

Removed repeated HTML artifacts from scraped data.

Referential Integrity

Deduplication strictly enforced via generated primary key.

7. DATA STORAGE

Relational Architecture



Relational Model

Structured row-column schema.



SQLite Engine

Lightweight analytical storage layer.

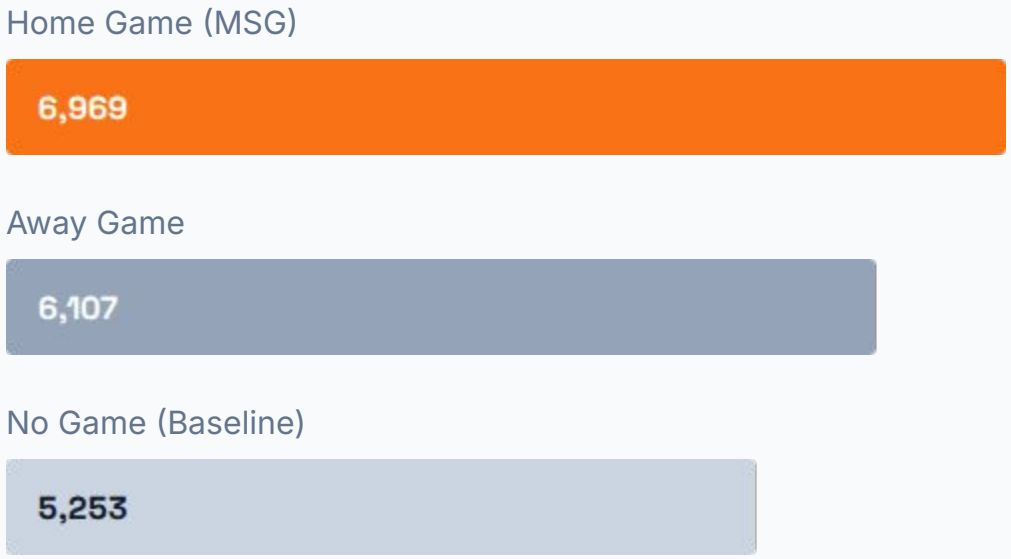


Incremental Updates

Seamless new record integration.

8.1 VALIDATION

The ”Madison Square Garden” Surge



+32%

Hourly trip increase.

Validating event-driven demand anomalies.

8.2 VALIDATION & 8.3 EDA

Meteorological Impact

Precipitation Intensity	Average Trips
Clear	**5,445**
Light Rain	4,350
Moderate Rain	3,667
Heavy Rain	4,239

Primary Driver

0.42

Pearson correlation (ρ) verifies temperature as the strongest demand driver.

9. CONCLUSIONS

Ready for Machine Learning.

Heterogeneous raw data successfully unified into a high-quality, reconciled Analytical Base Table.

Image Sources



https://www.nyc.gov/html/dot/images/pr2024/broadway-east-17th-to-east-21st-st.jpg

Source: www.nyc.gov



https://www.eagleview.com/wp-content/uploads/2026/02/02.09.26-EV-Blog-feature-image.jpg

Source: www.eagleview.com